① Smart Home Automation System Using Intelligent Agents

An intelligent agent is a system that observes its environment through sensors and take actions to work. In a smart home, sensors detect motion, temperature and time. while the system controls lights, air conditioning, and security devices.

Agent type		
Simple reflex	Responds immediately to sensor inputs	No memory
Model based	Tracks occupancy and environmental conditions	Needs internal state
Goal based	Works toward comfort and security goals	Requires planning

Utility based	Balances comfort energy security	More complex

A hybrid model-based and utility based would be effective

The model-based component remembers occupancy patterns and user preferences

Simple reflex rules provides ~~immedio~~ notification to emergencies.

② Robot Navigation in an Unknown Maze Using UCS and IDS

A robot is an unknown maze must explore the environment and find a path to exit without prior knowledge of the layout.

① Uniform Cost Search (UCS)

- UCS expands the node with the lowest total path cost

① Start from the initial position
② Add reachable to a queue
③ Select the position with the lowest cost
④ Continue exploring until the exit is reached
⑤ Return the lowest-cost path

① Iterative Deepening Search (IDS)

IDS repeatedly perfoms depth-limited search, increasing the depth limit after each unsucessful attempt.

① Begin with a depth limit of zero
② Search paths up to current limit
③ If the exit is not found, increase the limit
④ Repeat until the exit is reached

The choice of agent and search algorithm affects how efficiently a robot explores an unknown maze